

● Data Cleaning and Reporting Automation Using Excel

Objective

The objective of this project is to automate data cleaning and reporting workflows using Google Sheets/Excel. The project focuses on identifying missing values, removing duplicate records, correcting inconsistent data, and generating automated reports and visualizations.

● Original Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	Order ID	Customer Name	City	Product	Sales Amount	Date							
2	1001	Ravi	Hyderabad	Laptop	55000	01-01-2025							
3	1002	Sita	hyderabad	Mobile	20000	02-01-2025							
4	1003	Arjun	Chennai	Tablet		03-01-2025							
5	1004	Priya	Bangalore	Laptop	58000	04-01-2025							
6	1005	Kiran	Hyderabad	Mobile		05-01-2025							
7	1006	Anu	Chennai	Headphones	3000	06-01-2025							
8	1007	Raj	BANGALORE	Tablet	16000	07-01-2025							
9	1008	Meena	Hyderabad	Laptop	57000	08-01-2025							
10	1009	Sai	Chennai	Mobile	21000	09-01-2025							
11	1010	Divya	Bangalore	Headphones	3500	10-01-2025							
12	1011	Lakshmi	Hyderabad	Laptop	60000	11-01-2025							
13	1012	Arun	Chennai	Tablet	14500	12-01-2025							
14	1013	Sneha	Hyderabad	Mobile	24000	13-01-2025							
15	1014	Rahul	Bangalore	Laptop	53000	14-01-2025							
16	1015	Kavya	Hyderabad	Tablet	17000	15-01-2025							
17	1016	Mohan	Chennai	Mobile	25000	16-01-2025							
18	1017	Pooja	Hyderabad	Laptop		17-01-2025							
19	1018	Nikhil	Bangalore	Headphones	3200	18-01-2025							
20	1019	Deepa	Hyderabad	Mobile	23000	19-01-2025							
21	1020	Ramesh	Chennai	Tablet	18000	20-01-2025							
22	1021	Ravi	Hyderabad	Laptop	55000	01-01-2025							
23	1022	Sita	hyderabad	Mobile		22-01-2025							
24	1023	Arjun		Tablet	15500	23-01-2025							
25	1024	Priya	Bangalore	Laptop	58000	04-01-2025							
26	1025	Kiran	Hyderabad	Mobile	22000	25-01-2025							
27	1026	Anu	Chennai	Headphones	2900	26-01-2025							
28	1027	Raj	Bangalore	Tablet	16500	27-01-2025							

The original dataset contains sales transaction records including Order ID, Customer Name, City, Product, Sales Amount, and Date. The dataset contains data quality issues such as missing values, duplicate records, and inconsistent city names that require cleaning before analysis.

- **Missing Values Detection**

C55:D57 fx

Temporary filter 1 Range: A1:F51

	A	B	C	D	E	F	G	H	I
1	Order ID	Customer Name	City	Product	Sales Amount	Date			
4	1003	Arjun	Chennai	Tablet		03-01-2025			
6	1005	Kiran	Hyderabad	Mobile		05-01-2025			
18	1017	Pooja	Hyderabad	Laptop		17-01-2025			
23	1022	Sita	Hyderabad	Mobile		22-01-2025			
32	1031	Lakshmi	Hyderabad	Laptop		31-01-2025			
47	1046	Farhan	Chennai	Headphones	3600	15-02-2025			
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Duplicate records were identified and removed using the Data Cleanup feature in Google Sheets. Removing duplicates improved data accuracy and prevented repeated records from affecting the analysis.

- **Data Cleaning Process**

	A	B	C	D	E	F	G
	Order ID	Customer Name	City	Product	Sales Amount	Date	
4	1003	Arjun	Chennai	Tablet	17000	03-01-2025	
5	1005	Kiran	Hyderabad	Mobile	22000	05-01-2025	
8	1038	Nikhil	Bangalore	Headphones	3300	07-02-2025	
3	1017	Pooja	Hyderabad	Laptop	60000	17-01-2025	
0	1022	Sita	Hyderabad	Mobile	22000	22-01-2025	
1	1029	Sai	Chennai	Mobile	21500	29-01-2025	
2	1030	Divya	Bangalore	Headphones	3400	30-01-2025	
9	1031	Lakshmi	Hyderabad	Laptop	61000	31-01-2025	
2							
3							
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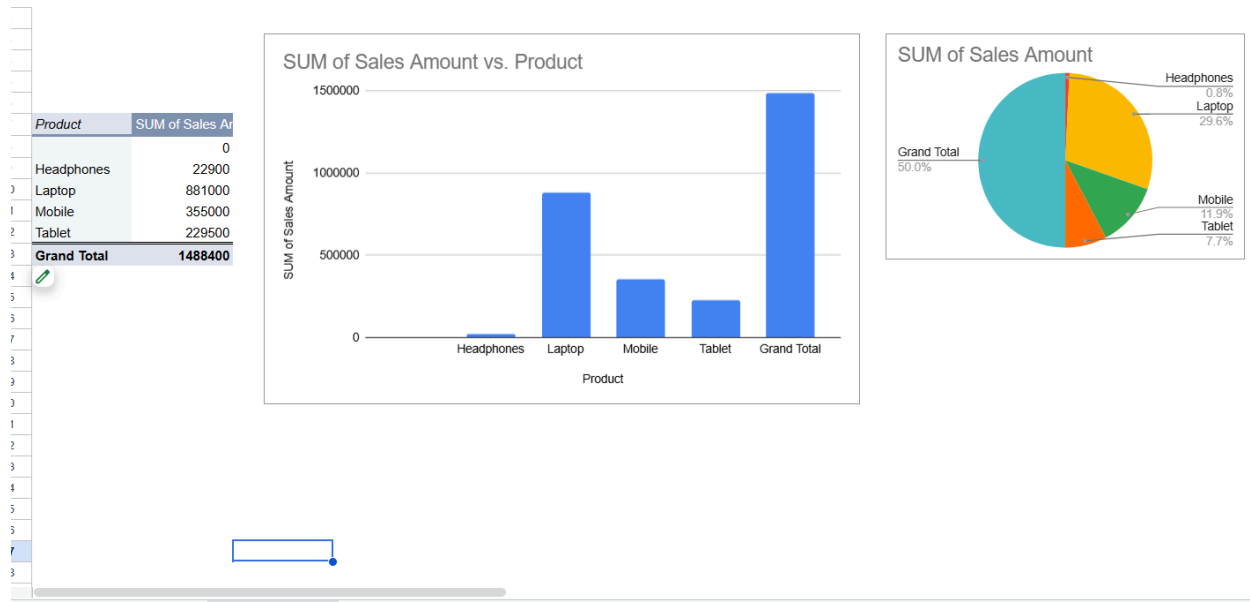
Missing Sales Amount values were filled with appropriate values, and the missing City value was corrected. Inconsistent city names such as "hyderabad" and "BANGALORE" were standardized to maintain consistency across the dataset.

● Cleaned Dataset

Order ID	Customer Name	City	Product	Sales Amount	Date						
1001	Ravi	Hyderabad	Laptop	55000	01-01-2025						
1002	Sita	hyderabad	Mobile	20000	02-01-2025						
1003	Arjun	Chennai	Tablet	17000	03-01-2025						
1004	Priya	Bangalore	Laptop	58000	04-01-2025						
1005	Kiran	Hyderabad	Mobile	22000	05-01-2025						
1006	Anu	Chennai	Headphones	3000	06-01-2025						
1007	Raj	BANGALORE	Tablet	16000	07-01-2025						
1008	Meena	Hyderabad	Laptop	57000	08-01-2025						
1009	Sai	Chennai	Mobile	21000	09-01-2025						
1010	Divya	Bangalore	Headphones	3500	10-01-2025						
1011	Lakshmi	Hyderabad	Laptop	60000	11-01-2025						
1012	Arun	Chennai	Tablet	14500	12-01-2025						
1013	Sneha	Hyderabad	Mobile	24000	13-01-2025						
1014	Rahul	Bangalore	Laptop	53000	14-01-2025						
1015	Kavya	Hyderabad	Tablet	17000	15-01-2025						
1016	Mohan	Chennai	Mobile	25000	16-01-2025						
1038	Nikhil	Bangalore	Headphones	3300	07-02-2025						
1018	Nikhil	Bangalore	Headphones	3200	18-01-2025						
1019	Deepa	Hyderabad	Mobile	23000	19-01-2025						
1020	Ramesh	Chennai	Tablet	18000	20-01-2025						
1021	Ravi	Hyderabad	Laptop	55000	01-01-2025						
1017	Pooja	Hyderabad	Laptop	60000	17-01-2025						
1023	Arjun		Tablet	15500	23-01-2025						
1024	Priya	Bangalore	Laptop	58000	04-01-2025						
1025	Kiran	Hyderabad	Mobile	22000	25-01-2025						
1026	Anu	Chennai	Headphones	2900	26-01-2025						
1027	Raj	Bangalore	Tablet	16500	27-01-2025						

The cleaned dataset contains complete and consistent information after handling missing values, removing duplicates, and correcting inconsistent city names. This dataset was used for further analysis and reporting.

Pivot Table Report



A Pivot Table was created to summarize total sales by product category. This automated report provides a quick overview of product performance and sales distribution.

Product-wise Sales Analysis (Column Chart)

The column chart visually represents the total sales generated by each product category. It helps identify the highest-performing products and compare sales across categories.

Sales Distribution by Product (Pie Chart)

The pie chart illustrates the percentage contribution of each product category to total sales. This visualization helps understand the overall sales distribution.

Results

- Missing values were successfully identified and corrected.
- Duplicate records were removed.
- Inconsistent city names were standardized.
- Automated reports were generated using Pivot Tables.
- Visual summaries were created using charts.

Conclusion

The dataset was successfully cleaned and processed using Google Sheets. Missing values, duplicate records, and inconsistent data were handled effectively. Automated reporting and visualization techniques improved data quality, reporting accuracy, and overall efficiency. This project demonstrates the importance of data preprocessing and reporting automation in business analytics and decision-making.